F70LA1 – Life Insurance Mathematics A, 2025/26
Chapter 3 – Life Tables

Read: Dickson, Hardy & Waters, Chapter 3.

3.1 Life tables

Life tables are tables of mortality-related statistics and probability values, calculated on a given population. They can be used to calculate the probability distribution of the future lifetime random variable, T_x.

3.1.1 Idea of constructing a life table

A basic life table shows how many people are expected to be alive at each age. The idea of constructing it, is as follows.

- Let 0 be the initial age (any age can be used, but let's keep it simple).
- Recruit a group of people who are all age 0. Suppose there are $\ell_0 > 0$ individuals in this original group.
  - The value ℓ_0 is called the *radix* (Latin word for 'root') of the table.
  - It is assumed that the future lifetimes of these people can be modelled by random variables that are independent of each other.
  - The number of people ℓ_0 should be large, so that the Law of Large Numbers holds. Let's assume that it holds with no error.
- Follow the group, counting how many of them survive to age 1 year, age 2 years, and so on. Denote these values by ℓ_1, ℓ_2, and so on.

The observed values $\ell_1, \ell_2, \ldots$ are then the expected number of people alive at each, as would be calculated if we knew explicitly the mortality distribution of the group.

In practice, constructors of life tables do not wait decades to calculate the table. More practical approaches are discussed in F79SU Survival Models.

Notation 3.1. *The actuarial notation for the number of survivors expected at age x is ℓ_x.*

3.1.2 Statistical underpinnings of a life table

Suppose there is a group of people who are all age 0. We assume that the future lifetimes of these people can be modelled by random variables that are independent of each other. We always make the assumption in what follows.

What can we say today – when there are $\ell_0 > 0$ of these people alive – about the number expected to survive to a future age $x > 0$?

- First consider one individual who is age 0 today. Suppose they have a probability ${}_xp_0$ of surviving to age $x > 0$. We can model whether they are alive or dead using a Bernoulli random variable, which takes value 1 if the individual survives to age x, and value 0 otherwise.
- For $\ell_0 > 0$ similar individuals, how many of them survive to age x? We can model this as ℓ_0 independent Bernoulli trials. We know this has a binomial distribution, with a probability ${}_xp_0$ of "success", where success is defined as survival to age x.
- Let the random variable $\mathcal{L}_x$ represent the number of survivors at age x. Then

$$\mathcal{L}_x \sim \text{Bin}\left(\ell_0, {}_xp_0\right).$$

- We know how to calculate the expected number of "successes" (i.e. survivors to age x) of a binomially-distributed random variable. It is

$$\mathbb{E}\left(\mathcal{L}_x\right) = \ell_0 \, {}_xp_0$$

- Using standard actuarial notation,

$$\ell_x := \mathbb{E}\left(\mathcal{L}_x\right).$$

  This links the construction of a life table with statistical theory.

Exercise 3.2. *Suppose there are $\ell_0 = 100,000$ newborns, each with a probability of 0.95 of surviving to age 50. How many are expected to be alive age 50?*

The number expected to be alive age 50 is $\ell_{50} = \ell_0 \cdot {}_{50}p_0 = 100,000 \times 0.95 = 95,000$.

Example 3.3. *Write down an expression in terms of $\mathcal{L}_{49}$ and $\mathcal{L}_{50}$ for the (random) number of deaths between ages 49 and 50.*

The number of deaths between ages 49 and 50 is $\mathcal{L}_{49} - \mathcal{L}_{50}$. This is the difference between the number of survivors at age 49 and the (smaller) number of survivors at age 50.

Exercise 3.4. *Write down an expression in terms of $\mathcal{L}_x$ and $\mathcal{L}_{x+1}$ for the (random) number of deaths between ages x and $x+1$.*

The number of deaths between ages x and $x+1$ is the random variable $\mathcal{L}_x - \mathcal{L}_{x+1}$.

Notation 3.5. *The actuarial notation for the number of deaths expected to occur between ages x and $x+1$ is*

$$\boxed{d_x := \mathbb{E}\left(\mathcal{L}_x - \mathcal{L}_{x+1}\right) = \mathbb{E}\left(\mathcal{L}_x\right) - \left(\mathcal{L}_{x+1}\right) = \ell_x - \ell_{x+1}}$$

and, for $t \geq 0$, the actuarial notation for the number of deaths expected to occur between ages x and $x+t$ is

$$\boxed{{}_td_x := \mathbb{E}\left(\mathcal{L}_x - \mathcal{L}_{x+t}\right) = \mathbb{E}\left(\mathcal{L}_x\right) - \left(\mathcal{L}_{x+t}\right) = \ell_x - \ell_{x+t}.}$$

3.2 Relationships between notation

Exercise 3.6. *Explain why*

$$\boxed{\begin{aligned} {}_tp_x &= \frac{\ell_{x+t}}{\ell_x}. \\ {}_tq_x &= \frac{\ell_x - \ell_{x+t}}{\ell_x}. \\ {}_{u|t}q_x &= \frac{\ell_{x+u} - \ell_{x+u+t}}{\ell_x}. \end{aligned}}$$

Exercise 3.7. *Explain why*

$$\boxed{d_x = \ell_x\, q_x.}$$

Exercise 3.8. *Explain in words why $\ell_x \geq \ell_{x+t}$ for $t \geq 0$.*

Exercise 3.9. *Explain in words why $\lim_{x\to\infty} \ell_x = 0$.*

3.3 Summary

- The set of values $\{\ell_x; x \geq 0\}$ is called the *life table*.
  - ℓ_x may be defined for all real $x \geq 0$.
  - But in practice, it is very common for a life table to be defined and/or tabulated at integer ages only.
- The value of ℓ_0 does not matter as it is only the ratios of the ℓ_x values that we use. For instance, ${}_tp_x = \ell_{x+t}/\ell_x$, i.e. the relationship is such that we can derive all that we need from the set $\{\ell_x; x \geq 0\}$.
- The value of ℓ_0 is called the *r*adix for the life table.
- The set of numbers $\{\ell_x; x = 0, 1, 2, \ldots\}$ defines the underlying survival distribution at integer ages only, starting from age 0. An assumption can be used to approximate the survival distribution between integer ages; these are studied in Section 3.5.

3.4 Examples of life tables

- **ELT15**
  - Based on the population of England and Wales.
  - ELT15 means 'English Life Table number 15'.
  - Calculated using mortality data from 1990–2.
  - Separate tables for males and females.

Table 1: Extracts from life tables ELT15 (Females) and ELT15 (Males).

	ELT15 (Females)	ELT15 (Males)	
x	l_x	l_x	x
65	87 093	79 293	65
66	85 875	77 353	66
67	84 567	75 256	67
⋮	⋮	⋮	⋮

Exercise 3.10. *Calculate p_{65}, q_{66} and ${}_{1|}q_{65}$ using the life table ELT15 (Females).*

$$p_{65} = \frac{\ell_{66}}{\ell_{65}} = \frac{85\,875}{87\,093} = 0.98601.$$

$$q_{66} = \frac{\ell_{66} - \ell_{67}}{\ell_{66}} = \frac{85\,875 - 84\,567}{85\,875} = 0.01523.$$

$${}_{1|}q_{65} = \frac{\ell_{66} - \ell_{67}}{\ell_{65}} = \frac{85\,875 - 84\,567}{87\,093} = 0.01502.$$

- **AM92**
  - Based on UK males who are whole life or endowment policyholders.
  - Calculated using mortality data from 1991–4.

Table 2: Extract from life table AM92.

	AM92	
x	l_x	x
60	9 287.2164	60
⋮	⋮	⋮
65	8 821.2612	65
66	8 695.6199	66
67	8 557.0118	67
⋮	⋮	⋮

Exercise 3.11. *Show that, using life table AM92 as the mortality basis, that*

$$p_{65} = 0.98576, \qquad q_{66} = 0.01594 \qquad \textit{and} \qquad {}_{1|}q_{65} = 0.01571.$$

$$p_{65} = \frac{\ell_{66}}{\ell_{65}} = \frac{8\,695.6199}{8\,821.2612} = 0.98576.$$

$$q_{66} = \frac{\ell_{66} - \ell_{67}}{\ell_{66}} = \frac{8\,695.6199 - 8\,557.0118}{8\,695.6199} = 0.01594.$$

$${}_{1|}q_{65} = \frac{\ell_{66} - \ell_{67}}{\ell_{65}} = \frac{8\,695.6199 - 8\,557.0118}{8\,821.2612} = 0.01571.$$

Exercise 3.12. *Using the AM92 table, show that the probability:*

(a) (60) survives to age 65 is 0.94983.

(b) (60) dies before age 67 is 0.07862.

(c) (60) dies between ages 65 and 67 is 0.02845.

(a) The probability (60) survives to age 65 is

$${}_5p_{60} = \frac{\ell_{65}}{\ell_{60}} = \frac{8\,821.2612}{9\,287.2164} = 0.94983.$$

(b) The probability (60) dies before age 67 is

$$ {}_7q_{60} = 1 - {}_7p_{60} = 1 - \frac{\ell_{67}}{\ell_{60}} = 1 - \frac{8\,557.0118}{9\,287.2164} = 0.07862. $$

(c) The probability (60) dies between ages 65 and 67 is

$$ {}_{5|2}q_{60} = \frac{\ell_{65} - \ell_{67}}{\ell_{60}} = \frac{8\,821.2612 - \ell_{8\,557.0118}}{9\,287.2164} = 0.02845. $$

Exercise 3.13. *Show that* $e_{95} = 0.516$ *years when using the following life table:*

x	l_x	d_x
95	539	331
96	208	146
97	62	54
98	8	8
99	0	–

All individuals are dead by age 99, which means that ${}_4p_{95} = {}_5p_{95} = \cdots = 0$. The curtate expectation of life from age 95 is

$$ \begin{aligned} e_{95} = \sum_{k=1}^{\infty} {}_kp_{95} = \sum_{k=1}^{3} {}_kp_{95} &= p_{95} + {}_2p_{95} + {}_3p_{95} \\ &= \frac{\ell_{96}}{\ell_{95}} + \frac{\ell_{97}}{\ell_{95}} + \frac{\ell_{98}}{\ell_{95}} \\ &= \frac{208}{539} + \frac{62}{539} + \frac{8}{539} \\ &= 0.516 \text{ years.} \end{aligned} $$

3.5 Fractional age assumptions

Life tables are typically only given for integer ages. They show us values of μ_x, ℓ_x and q_x at these ages, and sometimes other functions. However we may need these values at non-integer ages, for instance at age 25.25 years.

Suppose we want to calculate $\ell_{25.25}$, but we have only the values ℓ_{25} and ℓ_{26}. What can be done, to approximate the value of $\ell_{25.25}$ from these two known values? We know the value of $\ell_{25.25}$ must lie between them.

There are three well-known approximations used to calculate values at non-integer ages. They all do some kind of interpolation between the values at integer ages. Some may be more reasonable, given the overall trend in the $\{\ell_x; x\}$ values; but this can only be judged by studying the values of $\{\ell_x; x\}$.

A summary of the three approximations is given next.

- The *uniform distribution of deaths (UDD)* is explored in some detail below. It uses linear interpolation between the number of survivors at integer ages, i.e.

$$\text{For integer age } x \geq 0 \text{ and time } t \in (0,1), \qquad \ell_{x+t} = (1-t)\ell_x + t\ell_{x+1}.$$

- The *constant force of mortality* assumes exponential interpolation between integer ages, which can be expressed as linear interpolation in the logarithmic domain, i.e.

$$\text{For integer age } x \geq 0 \text{ and time } t \in (0,1), \qquad \ln\left(\ell_{x+t}\right) = (1-t)\ln\left(\ell_x\right) + t\ln\left(\ell_{x+1}\right).$$

- The *Balducci hypothesis* assumes for integer age $x \geq 0$ and time $t \in (0,1)$ that ${}_{1-t}q_{x+t} = (1-t)q_x$. We do not study the Balducci hypothesis further.

3.5.1 Uniform Distribution of Deaths (UDD)

The Uniform Distribution of Deaths (UDD) assumption between integer ages can be expressed in three different, but equivalent, ways. They are mean that deaths occurring uniformly between integer ages. The first expression of UDD assumes a linear interpolation between the number of survivors at integer ages.

Definition 3.14. *The Uniform Distribution of Deaths (UDD) assumption between integer ages linearly interpolates between the number of survivors at integer ages, i.e.*

$$\boxed{\textit{For integer age } x \geq 0 \textit{ and time } t \in (0,1), \qquad \ell_{x+t} = (1-t)\ell_x + t\ell_{x+1}.} \qquad \text{(UDD0)}$$

Example 3.15. *Using ELT15 (Females) and assuming UDD,*

$$\ell_{65.5} = (1-0.5)\ell_{65} + 0.5\ell_{66} = 0.5 \times 87\,093 + 0.5 \times 85\,875 = 86\,484.$$

Example 3.16. *Using ELT15 (Females) and assuming UDD, show that* $\ell_{65.2} = 86\,849.4$.

$$\ell_{65.2} = (1-0.2)\ell_{65} + 0.2\ell_{66} = 0.8 \times 87\,093 + 0.2 \times 85\,875 = 86\,849.4.$$

Exercise 3.17. *Show that* (UDD0) *can be re-written as the number of deaths occurring uniformly between integer ages, i.e. for integer age $x \geq 0$ and time $t \in (0,1)$,*

$$ {}_t d_x = t \cdot d_x. $$

Rearranging $\ell_{x+t} = (1-t)\ell_x + t\ell_{x+1}$ gives

$$ \ell_x - \ell_{x+t} = t\,(\ell_x - \ell_{x+1}) \quad \Rightarrow \quad {}_t d_x = t \cdot d_x. $$

The latter exercise leads into the next proposition.

Proposition 3.18. *The assumption* (UDD0) *is equivalent to:*

$$ \boxed{\text{For integer age } x \geq 0 \text{ and time } t \in (0,1), \qquad {}_t q_x = t \cdot q_x.} \tag{UDD1} $$

Proof. Exercise.

Fixing integer age $x \geq 0$ and time $t \in [0,1]$, and assume (UDD0) holds. Then

$$ {}_t q_x = \frac{\ell_x - \ell_{x+t}}{\ell_x} \overset{\text{(UDD0)}}{=} \frac{\ell_x - ((1-t)\ell_x + t\ell_{x+1})}{\ell_x} = t\frac{\ell_x - \ell_{x+1}}{\ell_x} = t \cdot q_x. $$

Next, assume (UDD1) holds. Then

$$ {}_t q_x = t \cdot q_x \quad \Rightarrow \quad \frac{\ell_x - \ell_{x+t}}{\ell_x} = t\frac{\ell_x - \ell_{x+1}}{\ell_x}, $$

i.e. $\ell_x - \ell_{x+t} = t\,(\ell_x - \ell_{x+1})$. Rearranging gives

$$ \ell_{x+t} = (1-t)\ell_x + t\ell_{x+1}. $$

□

Proposition 3.19. *The assumption* (UDD0) *is equivalent to:*

For integer age $x \geq 0$, define $R_x := T_x - K_x$. Then $R_x \sim U(0,1)$ and R_x is independent of K_x. (UDD2)

Proof. See pages 45-46 of Dickson, Hardy, Waters 'Actuarial Mathematics for Life Contingent Risks'. □

Lemma 3.20. *Let the age $x \geq 0$ be an integer and choose $t \in (0,1)$. Then assuming UDD between integer ages,*

$$ \boxed{\frac{d}{dt}\,{}_t q_x = q_x.} $$

Proof. It is easiest to use (UDD1). Then

$$ \frac{d}{dt}\,{}_t q_x \overset{\text{(UDD1)}}{=} \frac{d}{dt}\,(t q_x) = q_x. $$

□

Lemma 3.21. *Let the age $x \geq 0$ be an integer and choose $t \in [0,1)$. Then assuming UDD between integer ages,*

$$\boxed{q_x = {}_tp_x \cdot \mu_{x+t}.}$$

Proof.

$$\frac{d}{dt}{}_tq_x = \frac{d}{dt}(tq_x) = q_x.$$

□

Exercise 3.22. *Use ELT15 (Females) and assume UDD between integer ages to show the following.*

(a) ${}_{15}p_{70} = 0.47619$.

$${}_{15}p_{70} = \frac{\ell_{85}}{\ell_{70}} = \frac{38\,081}{79\,970} = 0.47619.$$

(b) ${}_{15.5}p_{70} = 0.45302$.

$${}_{15.5}p_{70} = \frac{\ell_{85.5}}{\ell_{70}} = \frac{0.5 \times \ell_{85} + 0.5 \times \ell_{86}}{\ell_{70}} = \frac{0.5(38\,081 + 34\,375)}{79\,970} = 0.45302.$$

(c) ${}_{15}p_{70.5} = 0.45803$.

$${}_{15}p_{70.5} = \frac{\ell_{85.5}}{\ell_{70.5}} = \frac{0.5 \times \ell_{85} + 0.5 \times \ell_{86}}{0.5 \times \ell_{70} + 0.5 \times \ell_{71}} = \frac{0.5(38\,081 + 34\,375)}{0.5(79\,970 + 78\,219)} = 0.45803.$$

Exercise 3.23. *There are 100 people now aged 40 years, of whom 19 are expected to die before age 41 years. Show that 9.5 of them are expected to die between ages $40\frac{3}{8}$ and $40\frac{7}{8}$, assuming UDD between integer ages.*

We are asked to calculate $\ell_{40\frac{3}{8}} - \ell_{40\frac{7}{8}}$ and verify that the difference has value 9.5. The given information is $\ell_{40} = 100$ and $d_{40} = 19$, which implies that $\ell_{41} = \ell_{40} - d_{40} = 81$.

Applying the UDD assumption,

$$\ell_{40\frac{3}{8}} = \frac{5}{8}\ell_{40} + \frac{3}{8}\ell_{41} = \frac{5}{8}(100) + \frac{3}{8}(81) = 92.875.$$

and

$$\ell_{40\frac{7}{8}} = \frac{1}{8}\ell_{40} + \frac{7}{8}\ell_{41} = \frac{1}{8}(100) + \frac{7}{8}(81) = 83.375.$$

Thus $\ell_{40\frac{3}{8}} - \ell_{40\frac{7}{8}} = 92.875 - 83.375 = 9.5$.

For integer age x and for $s, t \in [0,1]$ such that $s+t \leq 1$, the next proposition shows that ${}_tq_{x+s} \neq {}_tq_x$, since ${}_tq_x = t \cdot q_x$ by (UDD1).

Proposition 3.24. *Show that the UDD assumption implies that for integer age x and for $s, t \in [0,1]$ such that $s+t \leq 1$,*

$$\boxed{{}_tq_{x+s} = \frac{t \cdot q_x}{1 - s \cdot q_x}.}$$

Proof. See Tutorial 3. □

Exercise 3.25. *Given that $\ell_{75} = 100$, $\ell_{76} = 80$, show that ${}_{0.3}q_{75.5} = 0.06667$ using the UDD assumption for fractional ages.*

We can calculate ${}_{0.3}q_{75.5}$ by applying the result of Proposition 3.24, i.e.

$${}_{0.3}q_{75.5} = \frac{0.3 \times q_{75}}{1 - 0.5 \times q_{75}}.$$

As

$$q_{75} = \frac{\ell_{75} - \ell_{76}}{\ell_{75}} = \frac{100 - 80}{100} = 0.20,$$

then

$${}_{0.3}q_{75.5} = \frac{0.3 \times 0.20}{1 - 0.5 \times 0.20} = 0.06667.$$

3.5.2 Constant Force of Mortality

Definition 3.26. *The* constant force of mortality *assumption between integer ages:*

$$\textit{For integer age } x \geq 0 \textit{ and time } t \in (0,1), \qquad \mu_{x+t} = -\ln(p_x).$$

In other words, the force of mortality is a constant value $-\ln(p_x) > 0$ between integer ages. Note that the constant $-\ln(p_x)$ is not the force of mortality at age x but is an average of the true force of mortality between ages x and $x+1$.

Proposition 3.27. *The constant force of mortality assumption is equivalent to exponential interpolation between integer ages, i.e.*

$$\textit{For integer age } x \geq 0 \textit{ and time } t \in (0,1), \qquad \ln(\ell_{x+t}) = (1-t)\ln(\ell_x) + t\ln(\ell_{x+1}).$$

Proof. Exercise. Rearrange the exponential interpolation expression to get

$$\ell_{x+t} = \exp\left(\ln(\ell_x)^{1-t} + \ln(\ell_{x+1})^t\right) = \exp\left(\ln(\ell_x)^{1-t}\right\} \cdot \exp\{\ln(\ell_{x+1})^t\right) = (\ell_x)^{1-t} \cdot (\ell_{x+1})^t.$$

Dividing across by ℓ_x gives

$$\frac{\ell_{x+t}}{\ell_x} = \left(\frac{\ell_{x+1}}{\ell_x}\right)^t \quad \Leftrightarrow \quad {}_tp_x = (p_x)^t .$$

Recall ${}_tp_x = e^{-\int_0^t \mu_{x+s} ds}$. Taking logs and then differentiating with respect to t,

$$\ln({}_tp_x) = t \ln(p_x) \quad \Leftrightarrow \quad -\int_0^t \mu_{x+s} ds = t \ln(p_x) \quad \Leftrightarrow \quad \mu_{x+t} = -\ln(p_x) .$$

□

3.6 Life tables

National life tables are based on the mortality experience of the entire population of a country. They allow us to understand population trends, which informs public policy. For example, how many retired people will need to be paid a state pension? How many hospitals should there be?

Life insurance companies use life tables to price and value their products. Similarly, scheme actuaries of defined benefit pension funds use them to value the pension liabilities.

What should they use as a mortality basis for their insurance products and pension funds? Is the mortality experience of the entire population of a country suitable? No, due to both evidence from data and general reasoning.

General reasoning

- Imagine there is a pension fund for staff members of accountancy firm: these are mostly generally well-paid, highly-educated individuals who have excellent working conditions and can afford lots of high-quality food, private healthcare, good quality accommodation, who do regular exercise, and so on. They are unlikely to have the same mortality as an average person of their national population, comparing at each age.

- Imagine an individual buys a life annuity at retirement from an insurance company, i.e. they pay the insurance company a large lump sum (e.g. £100,000) in exchange for a regular income of £8,000 per annum for as long as they live. Only the richer individuals in society have such a large sum of money, and in turn it is likely that they had a well-paid job and, with access to money and its consequent benefits, are likely to live longer than the average person of their national population.

Evidence of data Insurance companies track the mortality experience of their policyholders. Their data also shows that their policyholders live longer, on average, than national data would suggest. Private pension funds find the same result.

You must read Sections 3.4-3.5 (page 49-54) on 'National life tables' and 'Survival models for life insurance policyholders' in Dickson, Hardy & Waters.

3.7 Life insurance underwriting

A life insurance company wants to sell a whole life insurance product to Leo, which would pay out £50,000 whenever he dies. He would pay level, monthly premiums. How much should they charge Leo? Should whole life insurance policyholders be charged the same rate?

The *premium rate* depends on both the size of the policy (in this case, the size is £50,000) and *rating factors*. Both the size of the policy and the rating factors indicate the level of risk. Rating factors could be:

- Health factors such as age, policyholder's own health history and their family health history, height, weight.
- Lifestyle factors such as smoking, alcohol consumption, high-risk leisure activities such as skydiving, scuba-diving.
- Demographic factors such as sex, occupation (compare the risk being a firefighter to a desk-based job).

The collection of the rating factor data is done through the underwriting process.

Definition 3.28. Underwriting *is the process used to assess the risk associated with insuring an individual or entity. It involves evaluating the likelihood of a claim and determining the appropriate premium and terms for the insurance policy. It is done by* underwriters*, a specialist occupation.*

The underwriter does the following:

- Risk assessment by collecting information on the relevant rating factors.
- Determine eligibility and premiums: On the basis of the risk assessment, the underwriter decides whether or not the customer can be sold the policy. Using premium rates calculated by the actuary, the underwriter will choose the premium rate to charge the customer. They may discuss with the actuary whether a rate different to those calculated should be used.
- Collaborates with the actuary to ensure that these processes are effective and that premium rates are appropriate.

Some of these tasks may be automated, meaning that the customer gets a very fast response from the company.

You must read Section 1.3.5 (pages 8-10) on 'Underwriting' in Dickson, Hardy & Waters.

3.8 Selection

For most customers, underwriting is a gate through which only relatively healthy lives can pass. The improved mortality that results is known as *selection*. Selection is a numerically significant effect and so we have to take account of it in our life tables.

You must read Section 3.6 (pages 54-56) on 'Life insurance underwriting' in Dickson, Hardy & Waters.

It is observed from data that the selection effect of lighter (i.e. lower) mortality wears off after some time. A new policyholder, Leo, is less likely to die than existing policyholders, who have held their policy for several years. As Leo holds his policy for longer, his *select mortality rate* gets closer to that of the longer-serving policyholders. This means that he experiences different mortality rates each year, keeping his age constant. Within a few years of taking out the policy, Leo's mortality is the same as theirs, called the *ultimate mortality*. The *select period* is the length of time for Leo's mortality to reach the ultimate mortality, starting from when he took out the policy.

There will be a select life table for the first year he holds the policy, another select life table for the second year, and so on, until he reaches the ultimate life table.

To calculate his mortality rate, we need:

- the age at which he bought his policy, denoted $[x]$, and
- his current age, denoted $[x]+s$. (This means that the policy duration is s years, i.e. it has been s years since Leo bought the policy.)

Given these two ages, we can see which life table to use today, and which one to use for each subsequent year.

3.9 Select and ultimate tables

Definition 3.29. *For age $x \geq 0$ and time $s \geq 0$, define the one-year select mortality probability*

$$q_{[x]+s} := \mathbb{P}\left[\textit{a life age } x+s\textit{, who took out their policy at age } x\textit{, dies before age } x+s+1\right].$$

The corresponding one-year survival rate is

$$p_{[x]+s} := 1 - q_{[x]+s}.$$

Define the expected number of survivors for the select tables,

> $\ell_{[x]+s}$ *is the expected number of survivors at age* $x+s$ *among those who took out their policy at age* x.

Other life table functions are defined similarly.

3.9.1 Example for three-year select tables

Suppose the select period is 3 years, i.e. after 3 years, the mortality reverts to the ultimate life table $\{\ell_x; x \geq 0\}$ so that

$$\ell_{[x]+3} = \ell_{x+3}, \qquad \ell_{[x]+4} = \ell_{x+4}, \qquad \ldots.$$

For someone who takes out their policy at age x and when the select period is still 3 years, the expected number of survivors at each age – used to calculate $q_{[x]+s}$ – is as shown in the table.

Age	x	$x+1$	$x+2$	$x+3$	$x+4$	$\ldots$
Select age	$[x]$	$[x]+1$	$[x]+2$	$[x]+3$	$[x]+4$	$\ldots$
Mortality	Select	Select	Select	Ultimate	Ultimate	$\ldots$
Expected number of survivors at age	$\ell_{[x]}$	$\ell_{[x]+1}$	$\ell_{[x]+2}$	ℓ_{x+3}	ℓ_{x+4}	$\ldots$
One-year survival probability	$p_{[x]} = \frac{\ell_{[x]+1}}{\ell_{[x]}}$	$p_{[x]+1} = \frac{\ell_{[x]+2}}{\ell_{[x]+1}}$	$p_{[x]+2} = \frac{\ell_{x+3}}{\ell_{[x]+2}}$	$p_{x+3} = \frac{\ell_{x+4}}{\ell_{x+3}}$	$p_{x+4} = \frac{\ell_{x+5}}{\ell_{x+4}}$	$\ldots$

Four columns are required to show a select life table which has a select period of three years: $\{\ell_{[x]}; x \geq 0\}$, $\{\ell_{[x]+1}; x \geq 0\}$, $\{\ell_{[x]+2}; x \geq 0\}$ and the ultimate $\{\ell_x; x \geq 0\}$.

ELT15 (Females)

x	l_x	d_x	q_x	μ_x	$\mathring{e}_x$	x
55	94 532	450	0.004 75	0.004 51	26.357	55
56	94 082	499	0.005 31	0.005 03	25.481	56
57	93 583	554	0.005 92	0.005 62	24.614	57
58	93 029	614	0.006 60	0.006 26	23.757	58
59	92 415	683	0.007 39	0.007 00	22.912	59
60	91 732	761	0.008 30	0.007 86	22.079	60
61	90 971	839	0.009 22	0.008 80	21.259	61
62	90 132	915	0.010 15	0.009 72	20.452	62
63	89 217	1007	0.011 29	0.010 74	19.657	63
64	88 210	1117	0.012 66	0.012 03	18.875	64
65	87 093	1218	0.013 99	0.013 42	18.111	65
66	85 875	1308	0.015 23	0.014 70	17.361	66
67	84 567	1417	0.016 76	0.016 09	16.621	67
68	83 150	1533	0.018 44	0.017 74	15.896	68
69	81 617	1647	0.020 17	0.019 49	15.185	69
70	79 970	1751	0.021 90	0.021 23	14.487	70
71	78 219	1876	0.023 99	0.023 11	13.800	71
72	76 343	2056	0.026 93	0.025 69	13.127	72
73	74 287	2239	0.030 14	0.028 97	12.476	73
74	72 048	2366	0.032 84	0.032 03	11.848	74
75	69 682	2487	0.035 69	0.034 80	11.234	75
76	67 195	2634	0.039 19	0.038 03	10.631	76
77	64 561	2812	0.043 56	0.042 14	10.044	77
78	61 749	2984	0.048 33	0.046 94	9.478	78
79	58 765	3158	0.053 73	0.052 28	8.934	79
80	55 607	3314	0.059 61	0.058 27	8.413	80
81	52 293	3435	0.065 68	0.064 64	7.914	81
82	48 858	3526	0.072 16	0.071 31	7.435	82
83	45 332	3596	0.079 33	0.078 61	6.974	83
84	41 736	3655	0.087 57	0.086 91	6.532	84
85	38 081	3706	0.097 31	0.096 74	6.111	85
86	34 375	3724	0.108 33	0.108 41	5.715	86
87	30 651	3634	0.118 59	0.120 52	5.349	87
88	27 017	3475	0.128 60	0.131 74	5.002	88
89	23 542	3330	0.141 46	0.144 62	4.667	89
90	20 212	3143	0.155 50	0.160 53	4.354	90
91	17 069	2903	0.170 06	0.177 51	4.065	91
92	14 166	2631	0.185 73	0.195 73	3.797	92
93	11 535	2321	0.201 26	0.214 98	3.551	93
94	9 214	2008	0.217 90	0.234 90	3.322	94
95	7 206	1702	0.236 19	0.257 32	3.112	95
96	5 504	1395	0.253 44	0.281 14	2.925	96
97	4 109	1102	0.268 20	0.302 67	2.754	97
98	3 007	853	0.283 52	0.322 41	2.588	98
99	2 154	653	0.303 31	0.346 28	2.422	99
100	1 501	488	0.324 89	0.376 71	2.269	100
101	1 013	350	0.345 62	0.408 87	2.133	101
102	663	240	0.361 86	0.437 69	2.011	102
103	423	161	0.379 92	0.462 73	1.887	103
104	262	105	0.400 45	0.493 00	1.758	104
105	157	68	0.436 18	0.537 29	1.621	105
106	89	41	0.459 94	0.599 08	1.518	106
107	48	23	0.483 89	0.637 85	1.425	107
108	25	13	0.507 91	0.683 88	1.338	108
109	12	6	0.531 90	0.731 91	1.257	109
110	6	3	0.555 74	0.781 81	1.183	110
111	3	2	0.579 32	0.833 37	1.114	111
112	1	1	0.602 55	0.886 29	1.050	112

Exercise 3.30. *A select life table has a select period of 3 years. Its ultimate mortality is equivalent to ELT15 (Females). For all integer ages $x \geq 65$,*

$$p_{[x]} = 0.999, \qquad p_{[x-1]+1} = 0.998, \qquad p_{[x-2]+2} = 0.997.$$

Show that the probability a life currently aged 70 will survive to age 75:

(a) Given that the life was select at age 67, is 0.87135.

A life who is currently aged 70 who was select at age 67 is currently aged $[67]+3$. However, as the select period is 3 years, they experience the ultimate mortality starting from age 70. Thus the probability is

$$ {}_5p_{[67]+3} = {}_5p_{70} = \frac{\ell_{75}}{\ell_{70}} = \frac{69\,682}{79\,970} = 0.87135. $$

(b) Given that the life was select at age 68, is 0.88819.

A life who is currently aged 70 who was select at age 68 is currently aged $[68]+2$. This means they are still within the select period at age 70. To calculate $\ell_{[68]+2}$, use the information given in the question, $p_{[x-2]+2} = 0.997$, with $x := 70$, to find

$$ 0.997 = p_{[68]+2} = \frac{\ell_{[68]+3}}{\ell_{[68]+2}} = \frac{\ell_{71}}{\ell_{[68]+2}} \qquad \Rightarrow \qquad \ell_{[68]+2} = \frac{\ell_{71}}{0.997} = \frac{78\,219}{0.997} = 78\,454.363. $$

As they experience the ultimate mortality starting from age 71,

$$ {}_5p_{[68]+2} = \frac{\ell_{75}}{\ell_{[68]+2}} = \frac{69\,682}{78\,454.363} = 0.88819. $$

(c) Given that the life was select at age 69, is 0.90819.

A life who is currently aged 70 who was select at age 69 is currently aged $[69]+1$. This means they are still within the select period at age 70. To calculate $\ell_{[69]+1}$, use the information given in the question, $p_{[x-1]+1} = 0.998$, with $x := 70$, to find

$$ 0.998 = p_{[69]+1} = \frac{\ell_{[69]+2}}{\ell_{[69]+1}} \qquad \Rightarrow \qquad \ell_{[69]+1} = \frac{\ell_{[69]+2}}{0.998}. $$

Calculating $\ell_{[69]+2}$ requires using $p_{[x-2]+2} = 0.997$, with $x := 71$,

$$ 0.997 = p_{[69]+2} = \frac{\ell_{[69]+3}}{\ell_{[69]+2}} = \frac{\ell_{72}}{\ell_{[69]+2}} \qquad \Rightarrow \qquad \ell_{[69]+2} = \frac{\ell_{72}}{0.997} = \frac{76\,343}{0.997} = 76\,572.718. $$

Hence $\ell_{[69]+1} = \frac{\ell_{[69]+2}}{0.998} = \frac{76\,572.718}{0.998} = 76\,726.170$.

As they experience the ultimate mortality starting from age 71,

$$ {}_5p_{[69]+1} = \frac{\ell_{75}}{\ell_{[69]+1}} = \frac{69\,682}{76\,726.170} = 0.90819. $$

(d) Given that the life is select at age 70, is 0.93239.

A life who is currently aged 70 who was select at age 70 is currently aged [70] *and clearly within the select period at age 70. To calculate* $\ell_{[70]}$*, use the information given in the question, to find*

$$0.999 = p_{[70]} = \frac{\ell_{[70]+1}}{\ell_{[70]}}.$$

Rearranging and repeatedly applying the information given in the question,

$$\ell_{[70]} = \frac{\ell_{[70]+1}}{0.999} = \frac{\ell_{[70]+2}}{(0.998)(0.999)} = \frac{\ell_{73}}{(0.997)(0.998)(0.999)} = \frac{74\,287}{(0.997)(0.998)(0.999)} = 74\,734.586.$$

As they experience the ultimate mortality starting from age 71,

$${}_5p_{[70]} = \frac{\ell_{75}}{\ell_{[70]}} = \frac{69\,682}{76\,572.718} = 0.93239.$$

Definition 3.31. *For age* $x \geq 0$ *and times* $s, t, u \geq 0$*, define the deferred mortality probability that a life age* $x + s$*, who took out their policy at age* x*, dies between ages* $x + s + t$ *and* $x + s + t + u$ *as*

$$\boxed{{}_{t|u}q_{[x]+s} := {}_tp_{[x]+s} \cdot {}_uq_{[x]+s+t}.}$$

Exercise 3.32. *Show that*

$${}_{t|u}q_{[x]+s} = {}_tp_{[x]+s} - {}_{t+u}p_{[x]+s}.$$

By the definition of ${}_{t|u}q_{[x]+s}$ and using the multiplicative property of survival probabilities,

$${}_{t|u}q_{[x]+s} = {}_tp_{[x]+s}\,{}_uq_{[x]+s+t} = {}_tp_{[x]+s}\left(1 - {}_up_{[x]+s+t}\right) = {}_tp_{[x]+s} - {}_{t+u}p_{[x]+s}.$$

Exercise 3.33. *(a) Derive an expression for* ${}_{1|6}q_{[65]}$ *in terms of* $\{\ell_x; x\}$*, assuming a select period of 2 years.*

(b) Hence, using life table AM92 as the mortality basis, show that ${}_{1|6}q_{[65]} = 0.11953$*.*

(a) By the definition of the deferred mortality probability,

$${}_{1|6}q_{[65]} = p_{[65]} \cdot {}_6q_{[65]+1} = \frac{\ell_{[65]+1}}{\ell_{[65]}} \cdot \frac{\ell_{[65]+1} - \ell_{[65]+7}}{\ell_{[65]+1}} = \frac{\ell_{[65]+1} - \ell_{72}}{\ell_{[65]}}.$$

(b) From life table AM92,

$${}_{1|6}q_{[65]} = \frac{8\,686.2016 - 7\,637.6208}{8\,772.7359} = 0.11953.$$

AM92

x	$l_{[x]}$	$l_{[x-1]+1}$	l_x	x
65	8 772.735 9	8 813.688 1	8 821.261 2	65
66	8 640.048 1	8 686.201 6	8 695.619 9	66
67	8 493.518 7	8 545.353 2	8 557.011 8	67
68	8 332.139 6	8 390.161 1	8 404.491 6	68
69	8 154.931 8	8 219.639 0	8 237.132 9	69
70	7 960.977 6	8 032.860 6	8 054.054 4	70
71	7 749.465 9	7 828.968 6	7 854.450 8	71
72	7 519.702 7	7 607.240 0	7 637.620 8	72
73	7 271.146 1	7 367.082 8	7 403.008 4	73
74	7 003.521 6	7 108.105 2	7 150.240 1	74
75	6 716.823 1	6 830.184 4	6 879.167 3	75
76	6 411.345 9	6 533.500 8	6 589.925 8	76
77	6 087.808 4	6 218.575 9	6 282.980 3	77
78	5 747.362 4	5 886.362 8	5 959.168 0	78
79	5 391.640 0	5 538.279 1	5 619.757 7	79
80	5 022.793 1	5 176.222 4	5 266.460 4	80
81	4 643.512 9	4 802.629 0	4 901.478 9	81
82	4 257.005 6	4 420.452 5	4 527.496 0	82
83	3 866.988 4	4 033.146 7	4 147.670 8	83
84	3 477.592 9	3 644.632 7	3 765.599 8	84
85	3 093.286 3	3 259.186 2	3 385.247 9	85
86	2 718.712 8	2 881.346 7	3 010.839 5	86
87	2 358.529 9	2 515.731 0	2 646.741 6	87
88	2 017.229 8	2 166.880 5	2 297.297 6	88
89	1 698.908 9	1 839.045 8	1 966.649 9	89
90	1 407.055 0	1 535.980 1	1 658.554 5	90
91		1 260.735 4	1 376.190 6	91
92			1 121.988 9	92
93			897.502 5	93
94			703.324 2	94
95			539.064 3	95
96			403.402 3	96
97			294.206 1	97
98			208.706 0	98
99			143.712 0	99
100			95.847 6	100
101			61.773 3	101
102			38.379 6	102
103			22.928 4	103
104			13.135 9	104
105			7.196 8	105
106			3.759 6	106
107			1.866 9	107
108			0.878 4	108
109			0.390 3	109
110			0.163 2	110
111			0.064 0	111
112			0.023 4	112
113			0.008 0	113
114			0.002 5	114
115			0.000 7	115
116			0.000 2	116
117			0.000 0	117
118			0.000 0	118
119			0.000 0	119
120			0.000 0	120